Predicting and Characterizing Student Inactivity in Online Courses

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https://drive.google.com/file/d/1rWxHho40zGuAiyESuSW3zOdqUzHg-WyG/view?usp=sharing

Motivation

Online Learning and Inactivity

- Online courses: flexible access to education
- High students' **inactivity rate**: learners disengage before completing courses
- **Early identification** of at-risk students could enable interventions
- **Goal:** understand and predict student inactivity

Research Questions

RQ1: Can machine learning predict, on a weekly basis, student inactivity in the next 30 days?

- **Task:** Supervised (Binary classification)
- **Goal:** On each week of the course, predict whether a student will disengage in the next 30 days.
- **Models:** Logistic Regression, Decision Tree, Gradient Boosting

RQ2: What engagement trajectories exist among online learners?

- **Task:** Unsupervised Learning (Clustering)
- **Goal:** Cluster students by engagement patterns.
- **Method:** K-Means Clustering

Data Source

Open University Learning Analytics dataset (OULAD)

- ~32,000 students
- Multiple online course modules
- Millions of interaction records
- Focus on one course starting from Oct 2013.

Dataset Files

- Student interaction logs
- Assessment submissions
- Demographic information
- Registration and dropout timing
- Data are aggregated differently for the prediction and clustering tasks.

1. Predicting Learners Inactivity

Variables Information

Unit of analysis:

- **id_student × week_t:** one student at a specific week (week *t*) in the course
- **32834** observations after preprocessing.

Target variable:

- inactivity_next_30d (binary)
- 1 if student unregisters or does not engage with the course in the next 30 days
- 0 otherwise

Features information (30 numerical, 6 categorical)

Temporal Feature: Weeks since course start

Engagement Features
- Total clicks (last 1, 2, 3, 4 weeks)
- Clicks by activity type (last 4 weeks)
- Engagement trend (slope, volatility)

Assessment Features (last 4 weeks)
- Number of submitted assessments
- Average scores
- Missed assessments

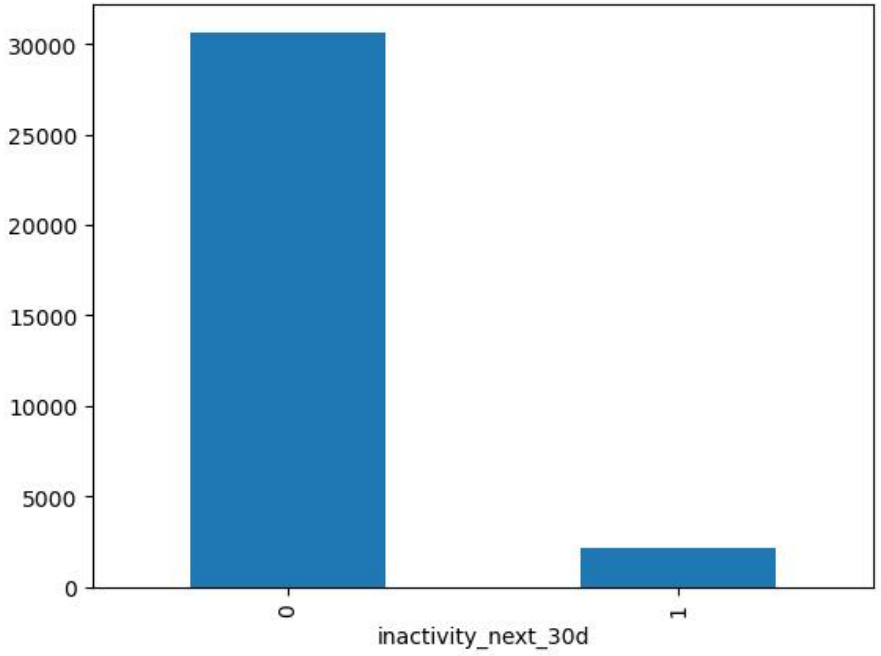
Demographic Features
- Gender, Age band, Education level, Region

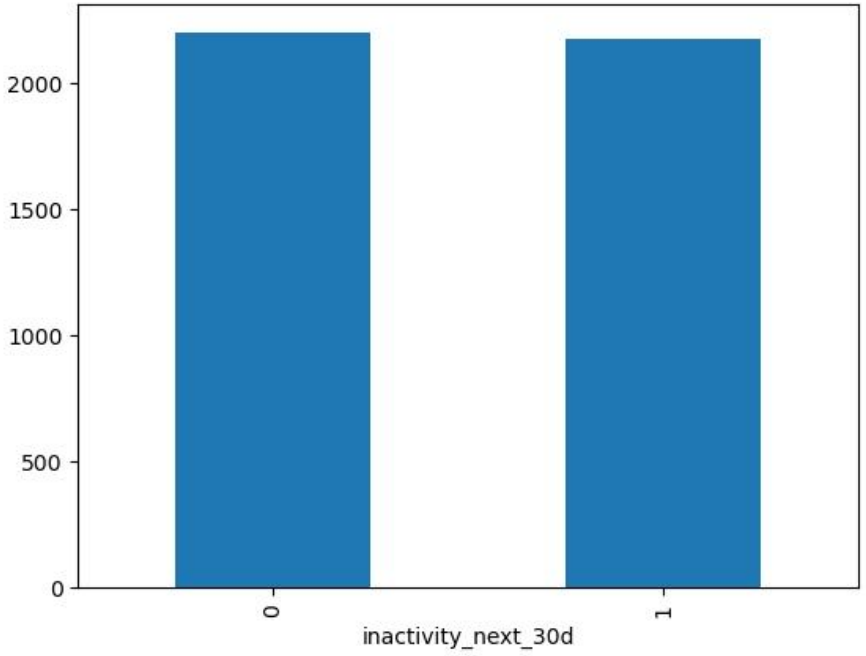
Exploratory Data Analysis

1. Address missing values
2. **Categorical features:** One-hot/ordinal encoding
3. **Numerical features:** outliers removal & MinMax scaling

Exploratory Data Analysis

Imbalanced class distribution: applied undersampling





Train-test split: split by “id_student” to prevent data leakage

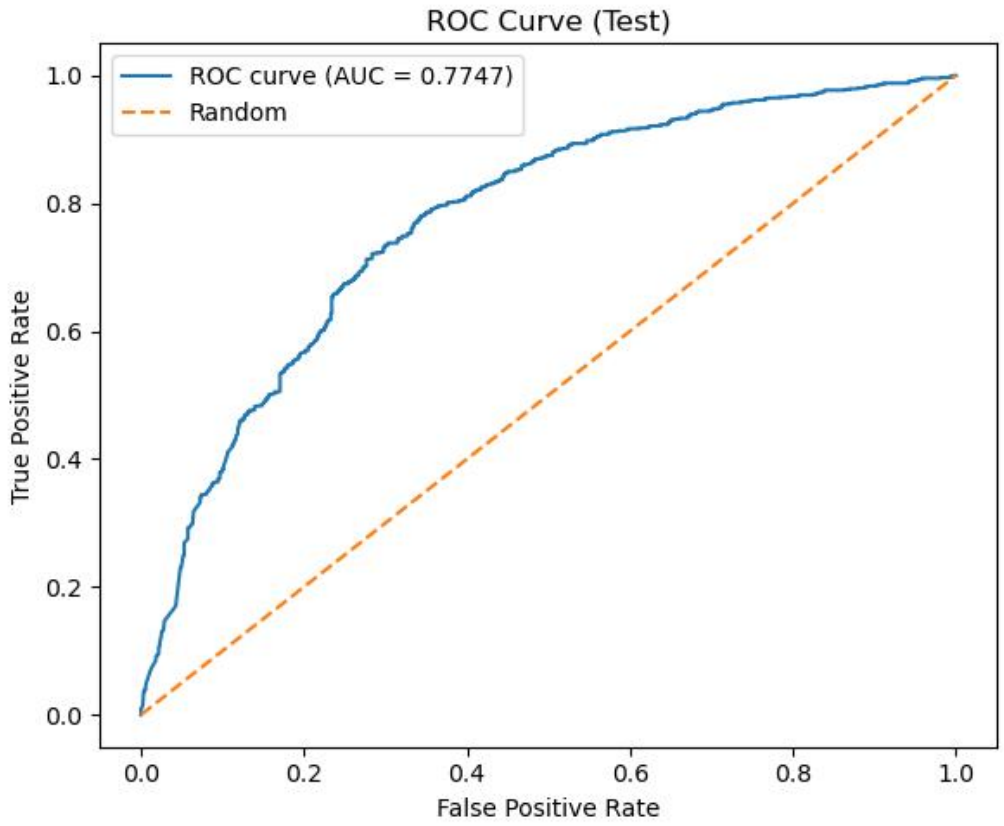
Model Exploration: Logistic Regression

Tuned using GridSearchCV

- Prevent leakage by using GroupKFold to split by “id_student”
- **Best hyperparameters:** Best params: {**'C'**: 0.1, **'class_weight':** 'balanced', **'penalty'**: 'l1', 'solver': 'liblinear'}

Moderate predictive performance, although more sensitive to detecting inactive cases (class 1).

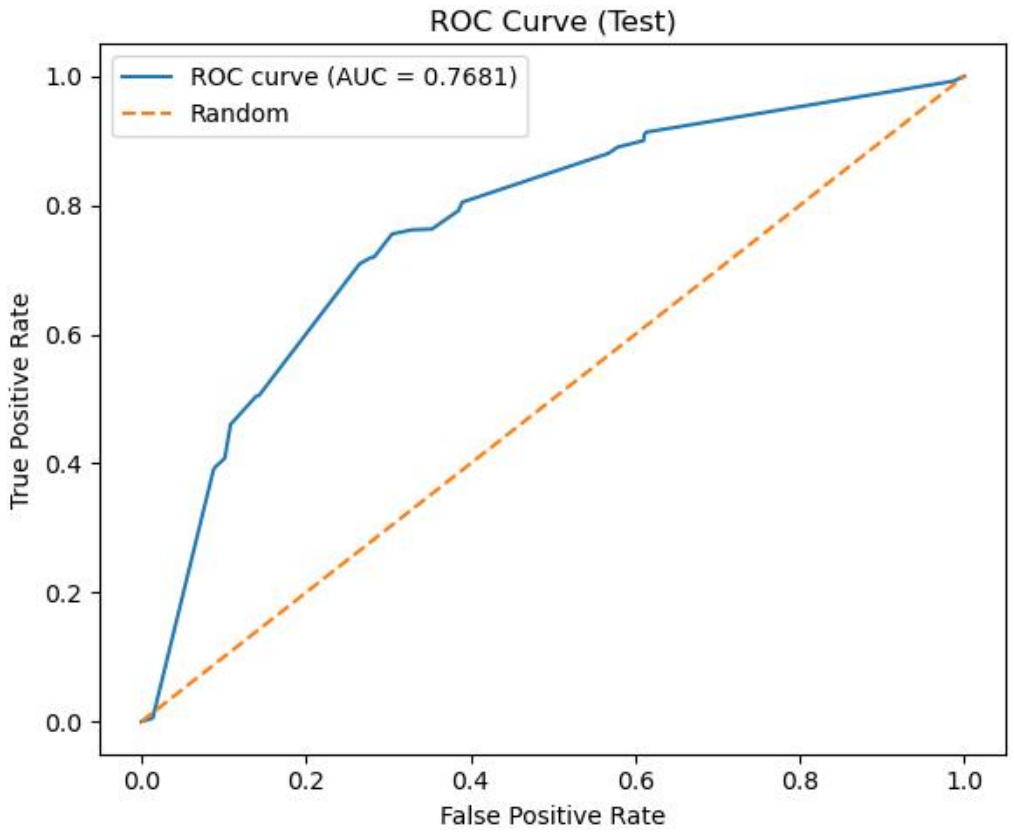
	precision	recall	f1-score	support
0	0.74	0.65	0.69	867
1	0.70	0.78	0.74	912
accuracy			0.72	1779
macro avg	0.72	0.72	0.72	1779
weighted avg	0.72	0.72	0.72	1779



Model Exploration: Decision Tree Classifier

- Baseline: overfit (train accuracy: 1.0; test: 0.66)
- Best hyperparameters: {'**class_weight**': None, '**criterion**': 'gini', **'max_depth'**: 5, 'max_features': None, **'min_samples_leaf'**: 5, **'min_samples_split'**: 20}
- Similar to LR: also better at predicting inactive cases (class 1).

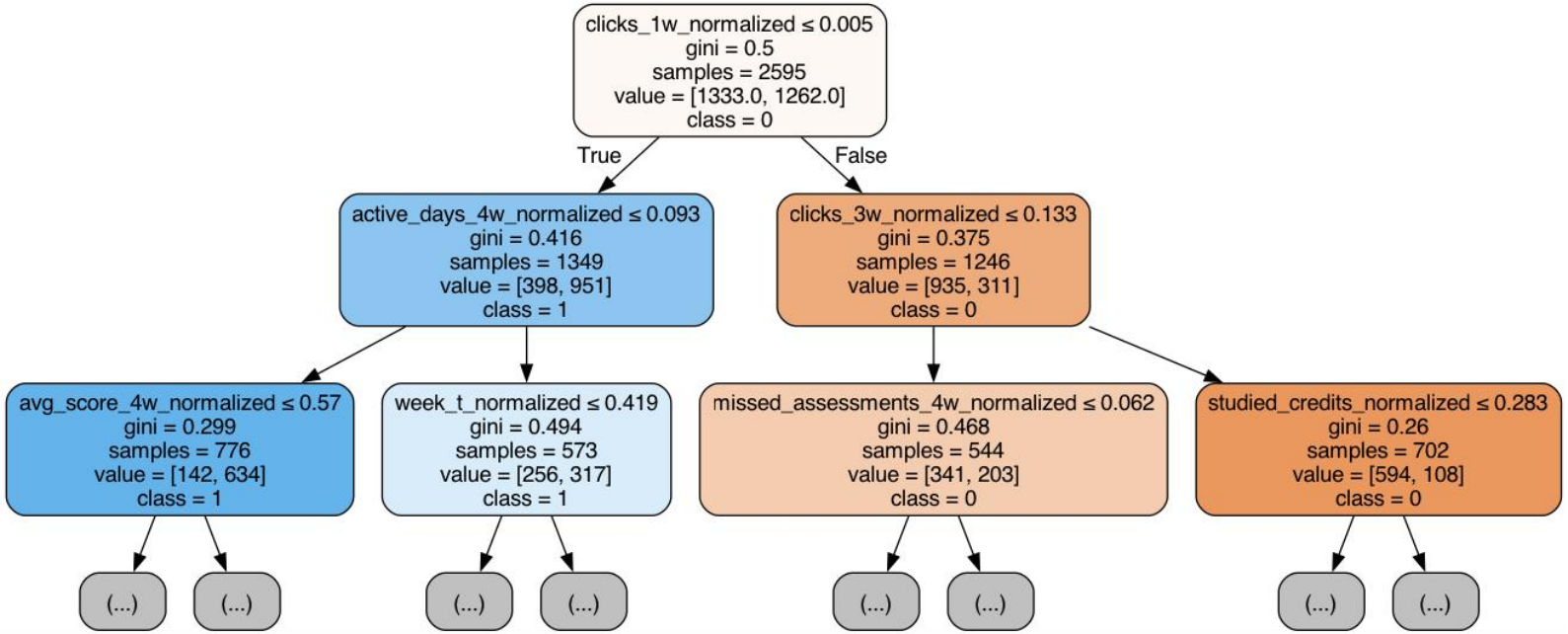
	precision	recall	f1-score	support
0	0.73	0.67	0.70	867
1	0.71	0.76	0.73	912
accuracy			0.72	1779
macro avg	0.72	0.72	0.72	1779
weighted avg	0.72	0.72	0.72	1779



Model Exploration: Decision Tree Classifier

Short-term behavioral pattern, rather than static characteristics, is the dominant driver of weekly inactivity prediction.

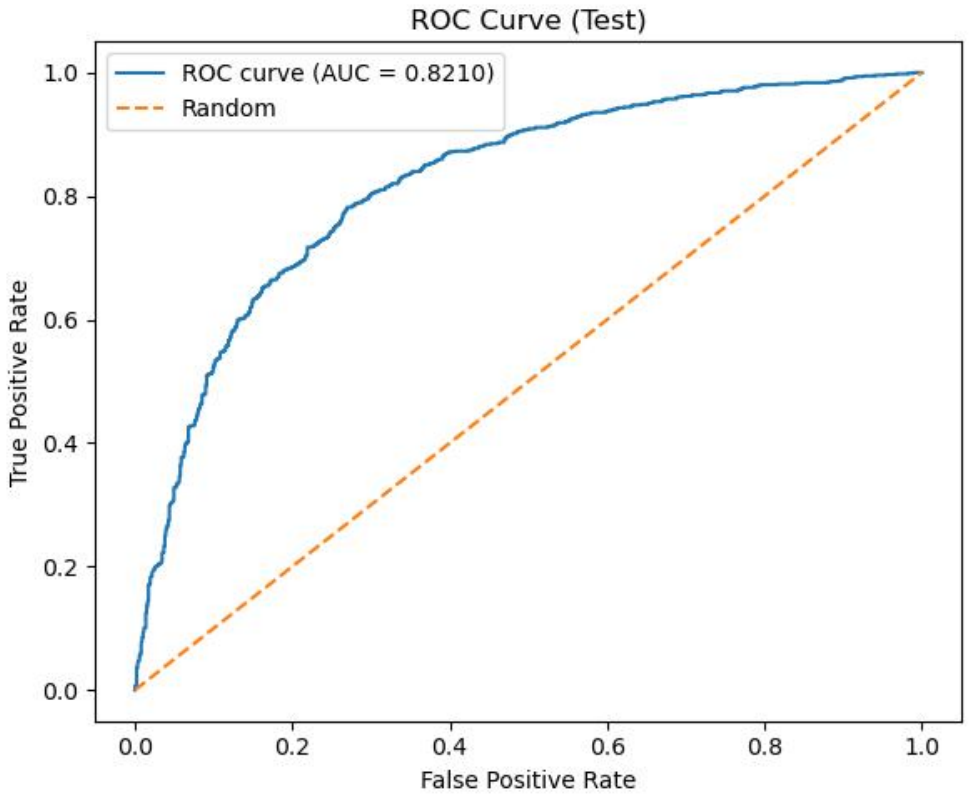
- Most important predictor: very recent engagement (`clicks_1w_normalized`)
- Subsequent splits: short-term engagement patterns such as `active_days_4w_normalized`, `clicks_3w_normalized`, and homepage interactions



Model Exploration: Gradient Boosting

- Best hyperparameters: {**'learning_rate'**: 0.05, **'max_depth'**: 3, **'max_features'**: None, '**min_samples_leaf**': 1, '**min_samples_split**': 2, '**n_estimators**': 200, **'subsample'**: 1.0}
- Precision and recall are well balanced across both classes.
- Provides better generalization than the 2 simpler models

	precision	recall	f1-score	support
0	0.74	0.74	0.74	867
1	0.76	0.75	0.75	912
accuracy			0.75	1779
macro avg	0.75	0.75	0.75	1779
weighted avg	0.75	0.75	0.75	1779



Model Comparison

- **Gradient Boosting:** best performance
  - Highest Accuracy (0.75) and ROC-AUC (0.82)
  - A well-balanced precision and recall rate
- **Logistic Regression & Decision Tree**
  - Slightly lower accuracy and ROC-AUC.
  - Slightly better at identifying **inactive** than active students (higher recall for Class 1)

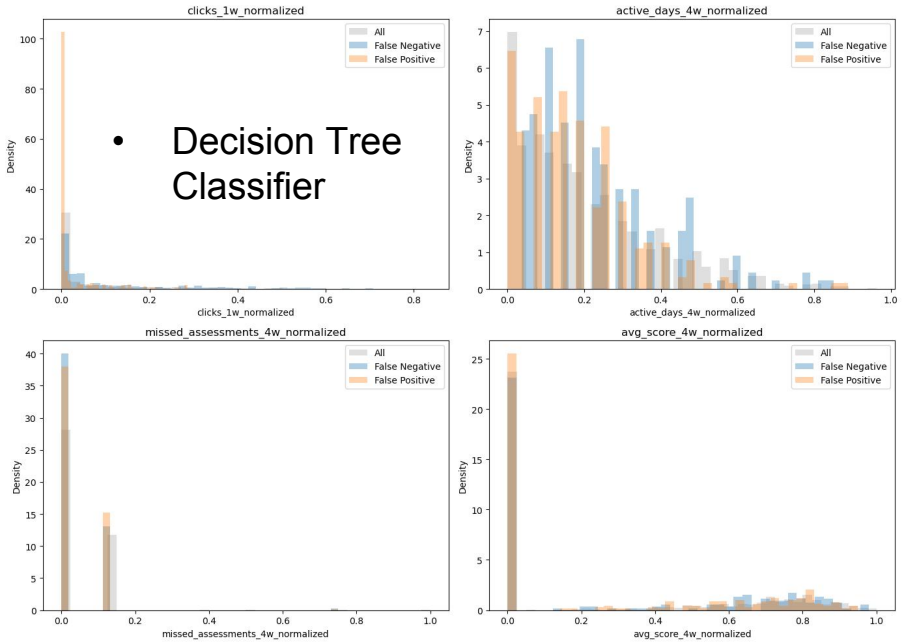
Why Gradient Boosting:

- Combines many trees sequentially to correct previous errors
- Captures nonlinear engagement patterns
- Controls overfitting through shallow trees and learning rates

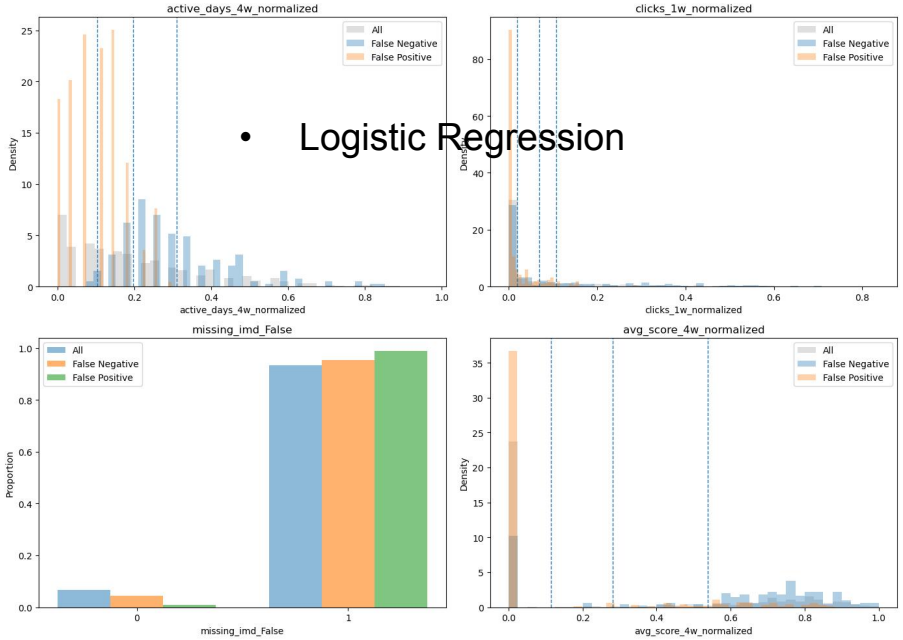
Model	Accuracy	Precision (Class 0)	Recall (Class 0)	Precision (Class 1)	Recall (Class 1)	ROC-AUC
Logistic Regression	0.72	0.74	0.65	0.70	0.78	0.77
Decision Tree	0.72	0.73	0.67	0.71	0.76	0.77
Gradient Boosting	0.75	0.74	0.74	0.76	0.75	0.82

Error Analysis:

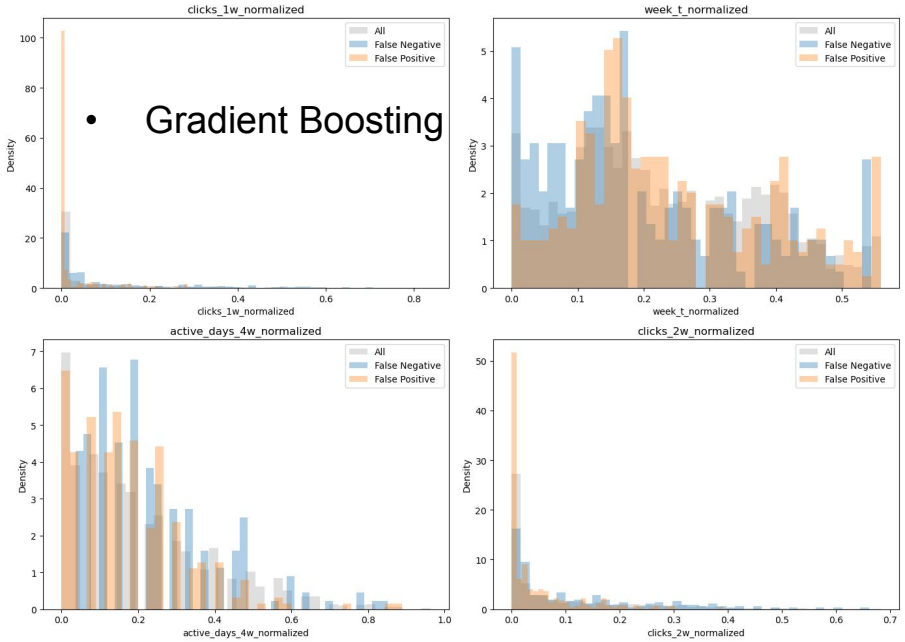
- Obtain the 4 most important features of each of the three models
- Compared features distribution of **False Positive** and **False Negative** cases against the entire sample.



- Decision Tree Classifier



- Logistic Regression



- Gradient Boosting

Error Analysis

- **Shared pattern across all three models:** Errors concentrate in **low-to-moderate engagement regions**, especially for `clicks_1w_normalized` and `active_days_4w_normalized`, where active and inactive students have **overlapping behavior** and are difficult to separate.

- **Gradient Boosting:** Errors are heavily concentrated in extremely low engagement values near zero, indicating difficulty distinguishing truly inactive students from temporarily inactive but still enrolled students.

2. Characterizing Engagement Patterns

Features Information:

Unit of Analysis: one student
- 2282 observations

Feature Categories (37 numerical features)
Overall Engagement Level
- Total course clicks
- Average weekly clicks and engagement variability
- Proportion of active weeks
- Longest inactivity streak
- Overall engagement trend over the course

Temporal Engagement Patterns
- Engagement in early, middle, and late stages of the course
- Change in engagement over time
- Difference between late-stage and early-stage activity

Engagement Timing
- Week of first activity
- Week of last activity
- Proportion of the course completed before last engagement

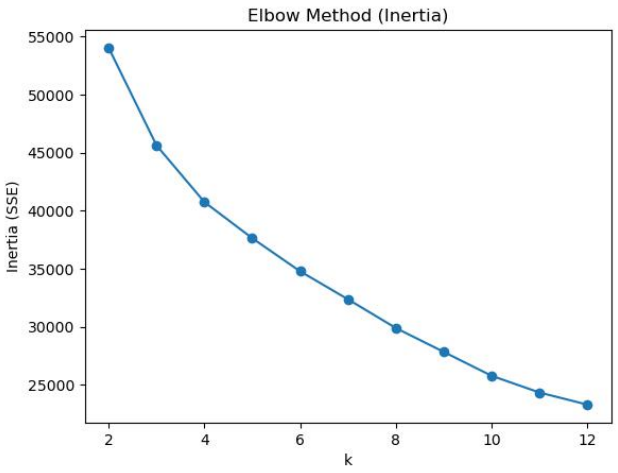
Activity-Type Composition
- Proportion of clicks across different learning activities

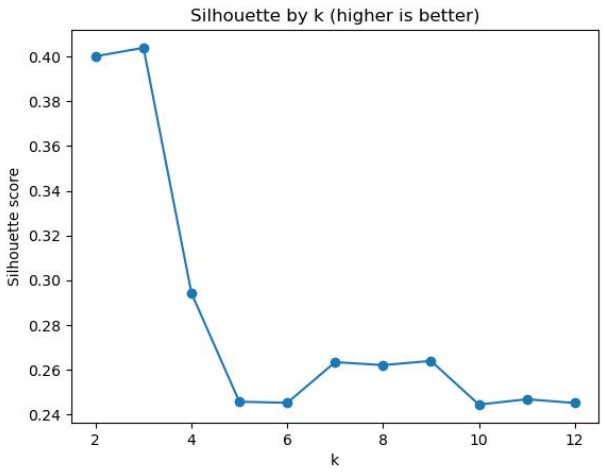
Assessment Behavior
- Number of assessments submitted, missed, or failed
- Assessment submission rate and missed rate
- Average assessment score
- Final exam participation and score

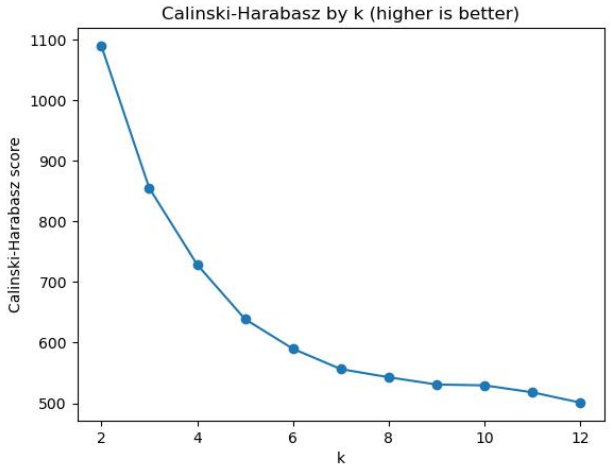
Applied K-Means clustering.

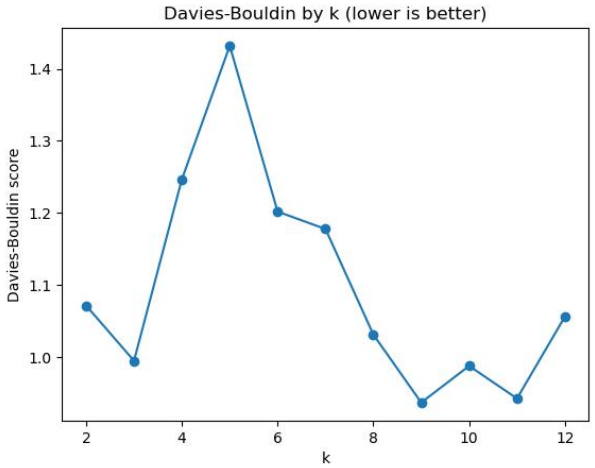
Best k: 3
clusters:

- cluster 0: 5
- cluster 1: 1252
- cluster 2: 1025









Clustering Results:

Cluster 0

	feature	z_score	centroid_original
0	pct_clicks_homepage	17.959	11.000
1	pct_clicks_subpage	16.081	7.183
2	pct_clicks_page	16.039	1.367
3	pct_clicks_resource	15.130	3.967
4	pct_clicks_oucontent	14.244	13.067
5	pct_clicks_forumng	14.047	7.450
6	pct_clicks_url	13.349	0.667
7	pct_clicks_ouwiki	6.559	0.133

Cluster 1

	feature	z_score	centroid_original
0	submission_rate	0.868	0.894
1	total_assess_submitted	0.868	11.625
2	total_assess_missed	-0.840	2.399
3	missed_rate	-0.840	0.185
4	longest_inactive_streak	-0.809	50.633
5	active_weeks_ratio	0.805	0.355
6	course_completion_ratio	0.767	0.416
7	week_of_last_activity	0.767	35.768

Cluster 2

	feature	z_score	centroid_original
0	submission_rate	-1.054	0.127
1	total_assess_submitted	-1.054	1.645
2	total_assess_missed	1.020	11.688
3	missed_rate	1.020	0.899
4	longest_inactive_streak	0.981	76.986
5	active_weeks_ratio	-0.976	0.078
6	week_of_last_activity	-0.931	10.528
7	course_completion_ratio	-0.931	0.122

Cluster	Engagement Pattern	Key Characteristics
Cluster 0	Passive Browsers	Mostly navigation clicks, minimal learning activity
Cluster 1	Highly Engaged Learners	High assessment submission, sustained engagement
Cluster 2	Early Disengaged Students	Few submissions, high missed rate, early dropout

Identifying Defining Features

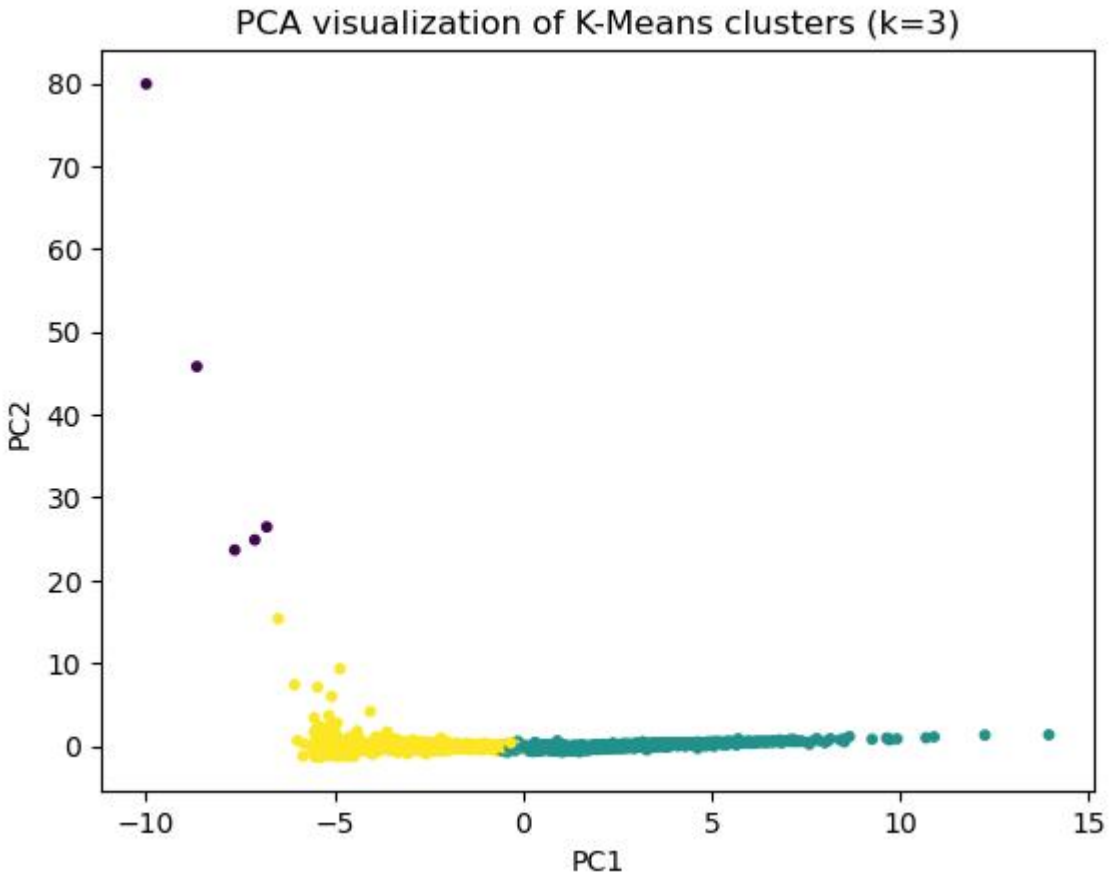
- Before clustering, all features were standardized so that each one has mean 0 and standard deviation 1.
- For each cluster, look at its **centroid values** in this standardized space by calling best_kmeans.cluster_centers_.
- **Calculate deviation of each cluster centriod from the feature average:** A large positive or negative value means that students in that cluster are far above or far below the overall average on that feature.
- Take the absolute value of these standardized differences and rank them from largest to smallest.
- The features with **the largest absolute standardized differences** are considered the defining features, because they are the ones where that cluster stands out the most compared to the overall population.

Visualization with PCA

Explained variance ratio:

- PC1: 0.390
- PC2: 0.143

Clear separation between engagement groups



3. Summary

Summary

Machine learning can help both predict student risk and understand engagement patterns.

- RQ1: Gradient Boosting achieves the best performance in predicting student engagement. Short-term engagement behaviors are most predictive of immediate disengagement.

- RQ2: Three distinct engagement trajectories: passive browsers, highly engaged learners, early disengaged students.

Thanks you!

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